tenstorrent

SEP

Security Processor

Beta release | Open-source RTL | Apache 2.0

HIGHLIGHTS

- VeeR EL2 RV32 processor with external interfaces for boot ROM, ICCM, DCCM, and scratch SRAM macros
- Integrated AES, HMAC, KMAC, OTBN, Adams Bridge, key-management, and entropy hardware
- OTP/eFuse-backed lifecycle controller with debug- and security-control outputs
- Filtered and remapped inbound and outbound system-fabric paths
- Secure DMA with inline hashing, quad-SPI host, watchdog, and mailboxes
- Open RTL, generated register collateral, firmware, and PyUVM verification

SYSTEM ROLE

The Security Processor (SEP) is the Root of Trust (RoT) of the chiplet with the highest security privilege. The SEP provides a physically isolated execution environment for highly sensitive security services such as secure boot, secure lifecycle control, device attestation, cryptography service and user authentication. The SEP is responsible for ensuring that the chiplet and overall packaged system composed of multiple chiplets remains secure from adversaries.

A primary chiplet's SEP provides system-level security management. Secondary chiplets may include their own SEP to enforce a local chiplet-owner security domain.

OVERVIEW

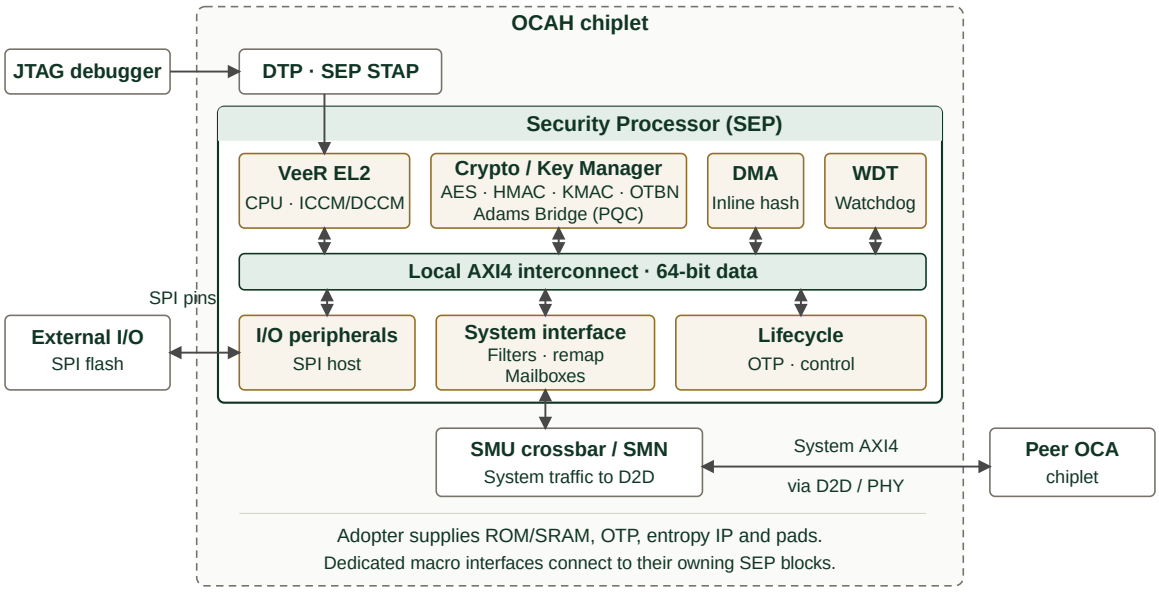
SEP combines a 32-bit VeeR EL2 processor, local memories, peripherals, and a crypto complex on a 64-bit local AXI4 fabric. Its lifecycle controller derives debug and security controls from eFuse-backed state. The 56-bit-address, 64-bit-data system AXI4 paths provide filtering and remapping. ROM, SRAM, OTP/eFuse, and entropy macros are supplied by the integrator. Boot firmware and crypto software are provided; their presence does not by itself establish a product secure-boot policy, certification, or complete system assurance case.

AT A GLANCE

Feature	Description
Processor	One VeeR EL2, RV32IMC plus configured bit-manipulation extensions
Crypto accelerators	Cipher AES; HMAC-SHA2; KMAC-SHA3; OTBN (programmable crypto engine); Adams Bridge (ML-DSA-87 and ML-KEM-1024)
TRNG	True Random Number Generator
PCR Vault	Deferred; not in the current RTL
Local memory apertures	256 KiB ICCM; 128 KiB DCCM; 64 KiB boot ROM; 256 KiB scratch SRAM
System / local AXI4	56-bit address, 64-bit data / 32-bit address, 64-bit data
Interrupt controller	255 sources: 43 internal and 212 external inputs
Fabric policy entries	32 outbound filters; 16 inbound filters
Mailboxes	8 mailboxes, depth 8 entries each
External entropy streams	3 AXI-Stream inputs, selected by default

The memory capacities and VeeR features are configurable at integration time. The properties of the associated technology macros, initialization images, timing, and physical characteristics depend on the selected instances and their configuration.

SYSTEM CONTEXT



© 2026 Tenstorrent USA, Inc. · Apache 2.0

CAPABILITIES

Processing and data movement

- VeeR EL2 CPU with instruction/data tightly coupled memories, programmable interrupt controller, timers, physical memory protection, and JTAG debug
- One secure-DMA instance supporting memory transfers; inline SHA2-256, SHA2-384, and SHA2-512 hashing; and the instantiated peripheral low-speed-I/O trigger path
- One single/quad-SPI host with one chip select; watchdog timer; and separate warm- and cold-reset scratch-register banks

Cryptographic and entropy hardware

- AES with 128-, 192-, and 256-bit keys; HMAC/SHA-2; KMAC/SHA-3/SHAKE; and programmable OTBN big-number computation
- Adams Bridge post-quantum accelerator instantiated with masking enabled in the reference configuration (ML-DSA-87 and ML-KEM-1024)
- Key Manager with private accelerator key paths and external ROM/SRAM ports
- Entropy source, CSRNG, EDN distribution, external-source selection, and a fabric-visible entropy pool

Lifecycle and fabric control

- eFuse sense/program interface, lifecycle-state logic, and lifecycle-derived controls for DTP and SMC integration
- Inbound access filtering; outbound filtering; alias remapping for CPU, DMA, and debug masters; and AP/STEE output-remap register banks
- Eight bidirectional mailbox paths and an external AXI extension port

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	System, reference, entropy-sample, and watchdog clocks; cold, debug, watchdog, scan, and software-controlled resets. The CPU reset is released after fuse sensing completes rather than as a separate controllable reset
System fabric	Inbound and outbound 56-address/64-data AXI4; SEP-to-SMC AXI4; 32-address/64-data external AXI extension
Debug and lifecycle	CPU JTAG; OTP-debug AXI4-Lite; lifecycle, debug-disable, security-disable, test-mode, and status signals
Memory macros	ICCM, DCCM, boot ROM, scratch SRAM, OTBN IMEM/DMEM, Adams Bridge SRAM, and Key Manager ROM/SRAM request/response interfaces
OTP and entropy	eFuse command and CSR interfaces; three external entropy AXI-Stream inputs, selected at reset, and one external-source AXI4-Lite control path
I/O and events	Quad-SPI pins, timer/software/external interrupts, watchdog reset request, mailbox interrupts, and error/status outputs
Build-time	Adams Bridge masking and SRAM latency, Key Manager read latency, vendor eFuse shim CSR block size, and security-disable token. The external entropy-stream count is fixed at three

INTEGRATION DEPENDENCIES

- Supply qualified ROM, SRAM, OTP/eFuse, and entropy IP matching the required interfaces, timing, integrity, reset, and images. Reset selects all three external entropy streams; their source IP is not supplied. Internal entropy sources require silicon validation.
- Replace the zero-valued default security-disable token during implementation and define lifecycle encodings, provisioning, debug authorization, recovery, and ownership policy for the complete product.
- Review firmware trust anchors, manifests, rollback, and update policy; source availability does not establish product approval.
- Integrate the inbound/outbound filters and remappers so configuration apertures cannot be exposed by an unsafe allow-list. Record the final address map and unmatched-access behavior.
- Provide clock, reset, CDC, scan, pad, interrupt, and fabric integration; connect DTP/SMC lifecycle handoffs and terminate unused ports.
- The integrator must qualify entropy and all cryptographic use in the target environment; the base system makes no certification, side-channel, fault-resistance, or implementation performance claims.

VERIFICATION AND MATURITY

The OCAH verification dashboard publishes the current verification and maturity status. The open-source cocotb/PyUVM environment runs against the SEP integration wrapper, combining the SEP RTL with memory macros and a generic eFuse model. It exercises CPU and ROM access, the address map, filters/remappers, DMA and SPI paths, mailboxes and interrupts, crypto modes and key paths, entropy distribution, lifecycle/eFuse behavior, reset isolation, and error handling. No-CPU tests use a CPU stub for direct bus stimulus; CPU firmware tests use the full VeeR core. Production Boot ROM firmware has a separate verification owner.

The behavioural OTP array does not prove physical OTP properties, and the Verilator build replaces the Adams Bridge key-register block with a compatibility stub. Published code coverage excludes the CPU subtree. No third-party algorithm or standards certification, production security assessment, silicon characterization, or portable frequency, area, power, latency, or throughput result is claimed.

DELIVERABLES

- SystemVerilog RTL, dependency filelists, and integration wrapper
- SystemRDL sources plus generated RTL, C headers, and documentation
- Architecture, memory-map, interface, and integration documentation
- Boot/DV firmware and cryptographic software source collateral
- cocotb/PyUVM environment, tests, checkers, and verification plan
- Apache License 2.0 repository distribution

RESOURCES

Documentation: docs.tenstorrent.com/tt-oca-harness
SEP source documentation: `hw/sys/sep/doc/`
RTL and registers: `hw/sys/sep/rtl/`, `hw/sys/sep/regs/`
Boot firmware: `hw/sys/sep/bootrom/`
Verification: `hw/sys/sep/dv/`
Integrator Guide: `doc/integrator/`
Project: github.com/tenstorrent/tt-oca-harness

TERMS

RoT: root of trust. TRNG: true random number generator. OTBN: OpenTitan big-number accelerator. HMAC: keyed-hash message authentication code. KMAC: KECCAK message authentication code. CSRNG: cryptographically secure random number generator. EDN: entropy distribution network. ICCM/DCCM: instruction/data closely coupled memory. OTP: one-time programmable.

DOCUMENT CONTROL

Version	Date	Author	Change

© 2026 Tenstorrent USA, Inc. · Apache 2.0